Chapter 4. Differentiable Functions

Derivatives, mean value theorems, L'Hôpital's rule, Darboux's theorem, Taylor's theorem

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4.1 Differentiability

Definition (Derivative)

$f : (a, b) \to \mathbb{R}$ is **differentiable at** x_0 if $f'(x_0) = \lim_{h\to 0} \frac{f(x_0+h)-f(x_0)}{h}$ exists.

Proposition 4.1 (Linear approximation)

f is differentiable at x_0 iff there are a number a and a function η with $\eta(h) \to 0 = \eta(0)$ such that

$$f(x_0 + h) = f(x_0) + ah + |h|\eta(h) \qquad \text{for small } h.$$

Then $a = f'(x_0)$. Equivalently, there is ϕ continuous at x_0 with $f(x) - f(x_0) = \phi(x)(x - x_0)$, and $\phi(x_0) = f'(x_0)$.

This form (the one that generalizes to several variables) gives the rules at once. A differentiable function is continuous. Sums, products and quotients obey the usual rules.

Proposition 4.2 (Chain rule)

If f is differentiable at x_0 and g at $y_0 = f(x_0)$, then $g \circ f$ is differentiable at x_0 with $(g \circ f)'(x_0) = g'(y_0)f'(x_0)$.

Proof. Write $f(x) - f(x_0) = \phi(x)(x - x_0)$ and $g(y) - g(y_0) = \psi(y)(y - y_0)$ with ϕ, ψ continuous at x_0, y_0. Then $g(f(x)) - g(f(x_0)) = \psi(f(x))\phi(x)(x - x_0)$, and $\psi(f(x))\phi(x) \to g'(y_0)f'(x_0)$. (The naive proof dividing by $f(x) - f(x_0)$ fails when this vanishes.) ■

Example 4.1

(a) $f(x) = x^2 \sin(1/x)$, $f(0) = 0$, is differentiable everywhere with $f'(0) = 0$, but $f'(x) = 2x \sin(1/x) - \cos(1/x)$ is discontinuous at 0.
(b) If f is differentiable at x_0 with $f'(x_0) \neq 0$ and invertible near x_0 with continuous inverse, then $(f^{-1})'(f(x_0)) = 1/f'(x_0)$.

4.2 Mean value theorems

If f has a local maximum or minimum at an interior point c where it is differentiable, then $f'(c) = 0$ (the difference quotients have opposite signs on the two sides).

Theorem 4.3 (Cauchy's mean value theorem)

If f, g are continuous on $[a, b]$ and differentiable on (a, b), there is $c \in (a, b)$ with

$$\{f(b) - f(a)\}g'(c) = \{g(b) - g(a)\}f'(c).$$

Proof. $h(x) = \{f(b) - f(a)\}g(x) - \{g(b) - g(a)\}f(x)$ satisfies $h(a) = h(b)$. If h is constant any c works; otherwise it attains a maximum or minimum at an interior point c, where $h'(c) = 0$ (Rolle's theorem). ■

With $g(x) = x$ this is the **mean value theorem**: $f(b) - f(a) = f'(c)(b - a)$. Hence a differentiable f on (a, b) is increasing, constant or decreasing according as $f' \geq 0$, $f' = 0$ or $f' \leq 0$; and $|f'| \leq L$ makes f Lipschitz with constant L.

Theorem 4.4 (L'Hôpital's rule)

Let f, g be differentiable on (a, b), $g' \neq 0$, and $\lim_{x\to a+} f(x) = \lim_{x\to a+} g(x) = 0$ (or $|g(x)| \to \infty$). If $f'(x)/g'(x) \to L$ as $x \to a+$, then $f(x)/g(x) \to L$.

Proof. In the $0/0$ case extend $f(a) = g(a) = 0$; for x near a, Cauchy's theorem gives $f(x)/g(x) = f'(c_x)/g'(c_x)$ with $a < c_x < x$. In the ∞ case fix y close to a with $|f'/g' - L| < \epsilon$ on (a, y) and write, for $a < x < y$, $\frac{f(x)-f(y)}{g(x)-g(y)} = \frac{f'(c)}{g'(c)}$; then $\frac{f(x)}{g(x)} = \frac{f'(c)}{g'(c)}(1 - \frac{g(y)}{g(x)}) + \frac{f(y)}{g(x)}$, and the last two corrections vanish as $x \to a$. ∎

Theorem 4.5 (Darboux)

If F is differentiable on $[a, b]$ and $F'(a) < \gamma < F'(b)$, then $F'(c) = \gamma$ for some $c \in (a, b)$.

Proof. $G(x) = F(x) - \gamma x$ has $G'(a) < 0 < G'(b)$, so G takes values below $G(a)$ just right of a and below $G(b)$ just left of b. Its minimum on $[a, b]$ is therefore attained at an interior point c, where $G'(c) = 0$. ∎

So derivatives have the intermediate value property although they need not be continuous: a function with a jump discontinuity (such as $\operatorname{sgn} x$) has no antiderivative on an interval.

4.3 Taylor's theorem

Theorem 4.6 (Taylor)

Let I be the closed interval with endpoints $a \neq b$, and suppose $f^{(n)}$ is continuous on I and $f^{(n+1)}$ exists in its interior. Then

$$f(b) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(b-a)^k + \frac{f^{(n+1)}(c)}{(n+1)!}(b-a)^{n+1}$$

for some c strictly between a and b (Lagrange form of the remainder).

Proof. Let $P(x) = \sum_{k\le n} f^{(k)}(x)(b-x)^k/k!$, so $P(b) = f(b)$ and, telescoping, $P'(x) = f^{(n+1)}(x)(b-x)^n/n!$. Apply Cauchy's mean value theorem to P and $g(x) = (b-x)^{n+1}$ on I: $\{P(b)-P(a)\}g'(c) = \{g(b) - g(a)\}P'(c)$, i.e. $\{f(b) - P(a)\}(n+1)(b-c)^n = (b-a)^{n+1}f^{(n+1)}(c)(b-c)^n/n!$. ∎

With $g(x) = b-x$ instead one gets the Cauchy form $\frac{f^{(n+1)}(c)}{n!}(b-c)^n(b-a)$; and if $f^{(n+1)}$ is Riemann integrable, the integral form $\frac{1}{n!}\int_a^b f^{(n+1)}(t)(b-t)^n dt$ follows from Chapter 5.

Proposition 4.7 (When the Taylor series converges)

Let f be C^∞ on an interval $J \ni a$. If there is M with $|f^{(k)}(x)| \le M$ for all $x \in J$ and all k, then the Taylor series of f at a converges to f on J.

Proof. The Lagrange remainder is at most $M|x-a|^{n+1}/(n+1)! \to 0$. ∎

Example 4.2

(a) $e^x = \sum x^k/k!$, $\sin x = \sum(-1)^k x^{2k+1}/(2k+1)!$ for all x, since all derivatives are bounded on bounded intervals.
(b) $f(x) = e^{-1/x^2}$ $(x \neq 0)$, $f(0) = 0$, is C^∞ with $f^{(k)}(0) = 0$ for all k (each derivative is a

polynomial in $1/x$ times e^{-1/x^2}). Its Taylor series at 0 is identically 0 and differs from f at every $x \neq 0$: C^∞ does not imply analytic.
(c) Second derivative test: if $f'(c) = 0$ and f'' is continuous with $f''(c) > 0$, then c is a strict local minimum, by Taylor's theorem with $n = 1$.